

APSC 2113 - Engineering Analysis I
Instructor: Professor Kartik Bulusu, MAE Dept., GWU
Diagnostic Test (LQ0) - Fall 2026

Questions

Q1. Find the derivative of

$$f(x) = 3x^4 - 5x^2 + 7.$$

- A. $12x^3 - 10x$ **(Correct Solution)**
- B. $12x^3 - 5x$
- C. $3x^3 - 10x$
- D. $12x^4 - 10x$

Explanation:

Applying the power rule $\frac{d}{dx}[x^n] = nx^{n-1}$ term-by-term yields $f'(x) = 3(4x^3) - 5(2x) + 0$. Simplifying this gives $12x^3 - 10x$.

Q2. Which of the following is the Taylor's series for $\sin x$ up to the x^5 term?

- A. $x + \frac{x^2}{2} + \frac{x^3}{6}$
- B. $x - \frac{x^3}{6} + \frac{x^5}{120}$ **(Correct Solution)**
- C. $1 - \frac{x^2}{2} + \frac{x^4}{24}$
- D. $x + \frac{x^3}{6} + \frac{x^5}{120}$

Explanation:

The Taylor series expansion centered at 0 for $\sin x$ is $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$. Evaluating the factorials gives $3! = 6$ and $5! = 120$, resulting in $x - \frac{x^3}{6} + \frac{x^5}{120}$.

Q3. Evaluate the limit using L'Hôpital's Rule

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}.$$

- A. 0
- B. 1 **(Correct Solution)**
- C. ∞
- D. Does not exist

Explanation:

Direct substitution yields the indeterminate form $\frac{0}{0}$, allowing L'Hôpital's Rule. Differentiating top and bottom gives $\lim_{x \rightarrow 0} \frac{\cos x}{1} = \cos(0) = 1$.

Q4. Simplify

$$\frac{x^5 x^{-2}}{x^3}.$$

- A. x^6
- B. 1 **(Correct Solution)**
- C. x^2

D. x^{-2}

Explanation:

Combine exponents in the numerator using exponent rules: $x^5 \cdot x^{-2} = x^{5-2} = x^3$.

Dividing numerator by denominator gives $\frac{x^3}{x^3} = x^{3-3} = x^0 = 1$.

Q5. Evaluate

$$\int x \ln(x) dx.$$

A. $\frac{x^2}{2} \ln x + \frac{x^2}{4} + C$

B. $\frac{x^2}{2} \ln x - \frac{x^2}{4} + C$ **(Correct Solution)**

C. $x \ln x - x + C$

D. $\frac{x^2}{2} \ln x - \frac{x}{2} + C$

Explanation:

Using integration by parts ($\int u dv = uv - \int v du$) with $u = \ln x$ and $dv = x dx$, we get $du = \frac{1}{x} dx$ and $v = \frac{x^2}{2}$.

This results in $\frac{x^2}{2} \ln x - \int \frac{x}{2} dx = \frac{x^2}{2} \ln x - \frac{x^2}{4} + C$.
